

Songs From The Wood

Concerts rock, right? But do you know just how much technology drives them?

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We're big rock fans here at *Digit*, and we're borderline obsessive-compulsive when it comes to the way our music sounds. We want guitars to gently weep, cymbals to crash just so, and drums that would rock the Casbah; "compromise" is a word whose use we don't condone. Perhaps that's why you will often find us trying new and strange ways to tweak and position our speakers—yes, even paltry two-speaker systems—in our quest for the ultimate music experience.

Sometime during said quest for perfect audio, we wondered how sound engineers at big concerts manage to keep thousands of people happily bouncing away to the sound of their favourite bands, and it boggled the mind.

And then we heard that Jethro Tull—one of the greatest rock bands of all time—was performing in Mumbai on January 31.

Hmm...how convenient.

"Just Don't Trip Over Anything"

So we endeared ourselves to the guys at SOUND.COM (see box SOUND.COM)—the sound engineers and audio equipment suppliers for the concert—and convinced them to let us tag along as they went about weaving their magic.

We inflicted ourselves upon Sunil Karanjikar, systems/mix engineer at SOUND.COM, who was our guide in this magical journey through a strange and wonderful mix of wires, musical instruments and mysterious odours at the Shanmukhananda Hall in Mumbai. Sunil is quite obviously the electronics geek—his face lit up as he told us about frequency responses, noise gates, power distributions and signal

processing.

On the morning of the concert, our heroes were already bleary-eyed from a night of setting up the huge speaker rigs, guided by the JBL Line Array Calculator—an Excel sheet provided by the manufacturer of the speakers—which suggests a starting point for their setup, based on the dimensions of the hall.

Men At Work

Shanmukhananda is an auditorium with three levels, and the first order of the day is to ensure that the sound reaches all parts of the hall—which is why the speakers arch upward. High sound frequencies are more directional than lower frequencies, so while they don't have to worry much about the bass, they do need to ensure that everyone gets the full effect of the highs.

To make sure the sound reaches its intended recipients, the gang uses a somewhat crude but useful tool—the laser pointer. "This way, we at least know that the speakers are pointing towards the *people* and not the wall or ceiling."

We wondered about the speakers lining the front of the stage. "Those are for the people in the front rows. The speaker arrays at the side will have their effect only from the fourth or fifth row, so without these, all that the front rows would hear is bass", Sunil explains.

Next step: getting the music to the back rows without having to blow out the ears of people in the front. You deserve the best sound even if you were a cheapskate with the tickets, so there is a set of speakers towards the end of the hall to make sure you get it.

Interlude—The Heart

In the thick of the action is the XTA DP428

